Numerical Simulation of One-Dimensional Wave Propagation in a Still Lake

Assessment of upwind and Lax–Wendroff schemes for the linearized shallow-water equations

1. Aim

The objective of this study is to compare first-order upwind and second-order Lax–Wendroff schemes for one-dimensional wave propagation. Their numerical diffusion, dispersion, stability and conservation properties are examined at different Courant–Friedrichs–Lewy (CFL) numbers.

2. Physical Model and Simulation Parameters

A small, initially localized Gaussian disturbance is introduced on the free surface of a still lake. The disturbance propagates in both directions through a periodic one-dimensional domain. The linearized shallow-water equations are used to describe the coupled variation of surface elevation and depth-averaged velocity.

Parameter	Value
Coordinate system	Cartesian
Grid points, nx	401
Grid points, ny	20
Initial disturbance amplitude	0.02
Final simulation time	2
CFL values specified	0.4, 0.8 and 1.1
Gravitational acceleration	9.81
Number of snapshots	180

3. Governing Equations

For small surface disturbances, the linearized shallow-water equations used in the source report are:

$$\partial \eta/\partial t + H\, \partial u/\partial x = 0$$

$$\partial u/\partial t + g\, \partial \eta/\partial x = 0$$

Here, η denotes the free-surface displacement, u is the depth-averaged horizontal velocity, H is the undisturbed water depth and g is gravitational acceleration. The wave speed is $c = \sqrt{(gH)}$, and the CFL number is defined as $c\Delta t/\Delta x$.

4. Boundary Conditions and Numerical Schemes

A periodic one-dimensional domain is used, with the initial condition represented by a localized Gaussian surface perturbation. The first-order upwind and second-order Lax–Wendroff methods are applied in conservative form to advance the linearized shallow-water system in time.

5. Parameters Used for Comparison

The simulations are compared at nominal CFL values of 0.4, 0.8 and 1.1. The main quantities examined are the final pulse amplitude and position, relative L2 error, error growth over time, visible phase displacement, spurious

oscillations and conservation of total surface elevation.

6. Numerical Results

The following table summarizes the numerical values reported at the final time t = 2. Values are approximate, as indicated in the source report.

Measured quantity	CFL = 0.4	CFL = 0.8	CFL = 1.1*
Pulse centres at t = 2	3.75 and 16.25	3.75 and 16.25	3.75 and 16.25
Analytical peak amplitude	0.0100	0.0100	0.0100
Upwind peak amplitude	0.0076	0.0090	0.0100
Upwind amplitude loss	24%	10%	Negligible
Lax–Wendroff peak amplitude	0.0100	0.0100	0.0100
Upwind relative L2 error	2.3 x 10^(-1)	9.5 x 10^(-2)	6 x 10^(-3)
Lax–Wendroff relative L2 error	2.4 x 10^(-2)	1.0 x 10^(-2)	6 x 10^(-4)

**The source labels CFL = 1.1 as nominal. Its conclusion notes that the effective CFL may have been limited to a value at or below one; therefore, this case should not be treated as proof of stability above the theoretical limit.*

7. Additional Observations

Observation	Reported behaviour
Error ratio (upwind / Lax–Wendroff)	Approximately 10 for all three CFL cases
Phase error (peak position)	No visible phase shift
Spurious oscillations	None visible
Error evolution	Smooth and monotonic growth
Mass conservation	Conserved by construction for both schemes

8. Discussion

The initial Gaussian disturbance divides into two pulses that travel in opposite directions. The reported pulse centres at t = 2 are approximately x = 3.75 and x = 16.25. Their displacement from the initial centre at x = 10 is about 6.25, giving a wave speed close to 3.1, consistent with the value expected from $c = \sqrt{(gH)}$ for the depth used in the source report.

The upwind method exhibits noticeable numerical diffusion. At CFL = 0.4, its peak amplitude decreases to approximately 0.0076, corresponding to a loss of about 24%. At CFL = 0.8, the loss reduces to around 10%, while the nominal CFL = 1.1 case shows negligible amplitude loss. The relative L2 error follows the same trend and decreases as the CFL number approaches unity.

Lax–Wendroff preserves the peak amplitude close to the analytical value of 0.0100 for all three cases. Its relative L2 error is approximately one order of magnitude lower than the upwind result at each CFL value. The supplied plots do not show visible phase displacement or spurious oscillations for this smooth Gaussian disturbance.

Both methods are conservative in the periodic domain because they are formulated in conservative form. Consequently, the reduction in the upwind pulse peak represents spreading of the same surface displacement over a wider region, rather than a loss of total mass.

The theoretical explicit stability limit stated in the source is $CFL \leq 1$. Although the nominal CFL = 1.1 plots do not show growing oscillations, the report suggests that the solver may have limited the actual time step. A run without such a limiter would be required to demonstrate the expected instability above the limit.

9. Conclusion

The one-dimensional linearized shallow-water problem was simulated using first-order upwind and second-order Lax–Wendroff schemes at nominal CFL values of 0.4, 0.8 and 1.1. The initial surface disturbance separates into two pulses that propagate in opposite directions, with their final positions agreeing with the expected wave speed.

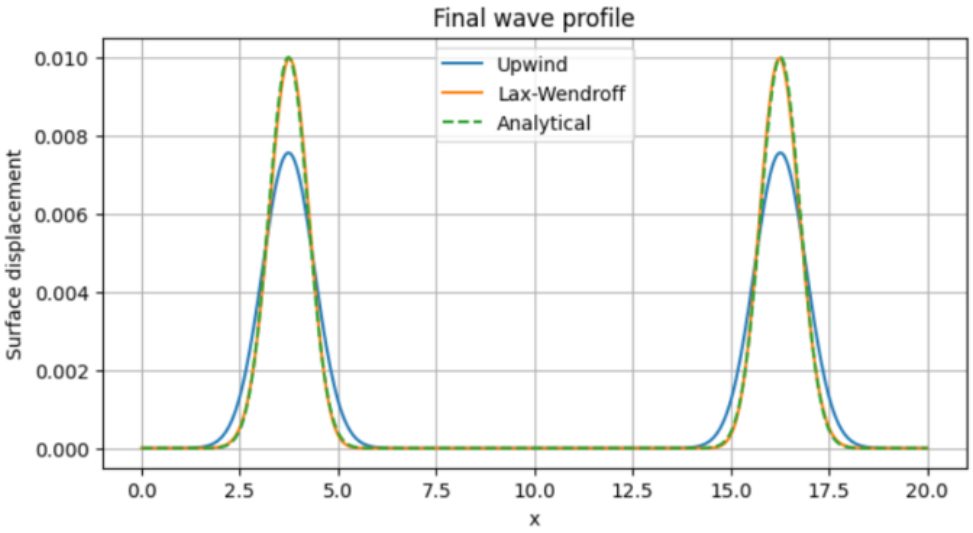
The results show that upwind is robust but introduces greater numerical diffusion, particularly at lower CFL values. Lax–Wendroff retains the pulse amplitude more accurately and produces lower relative L2 errors in all reported cases. Both schemes conserve total surface elevation in the periodic domain, and no visible phase error or oscillations are reported for the smooth initial pulse.

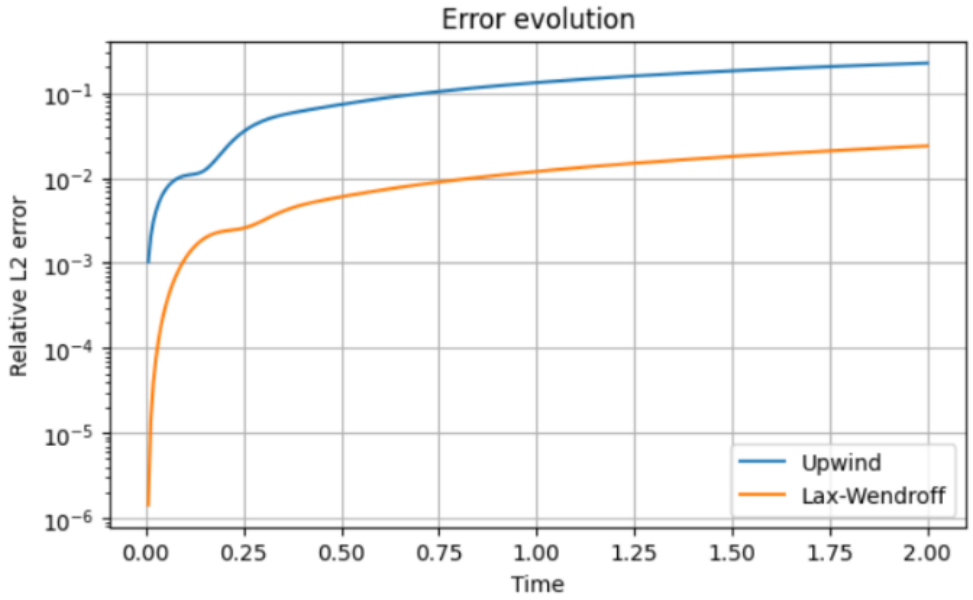
The nominal CFL = 1.1 case requires careful interpretation because the source indicates that the effective CFL may have been constrained by a time-step limiter. The stated theoretical stability condition remains CFL ≤ 1 for these explicit methods.

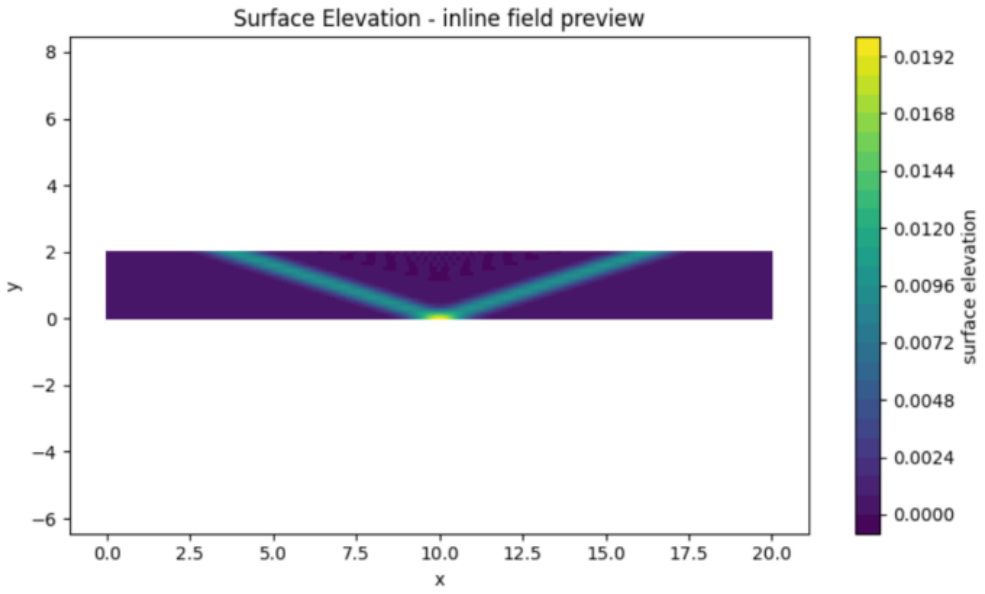
10. Original Simulation Figures

The original final wave profiles, error histories, surface-elevation fields and result tables are retained in the following pages.

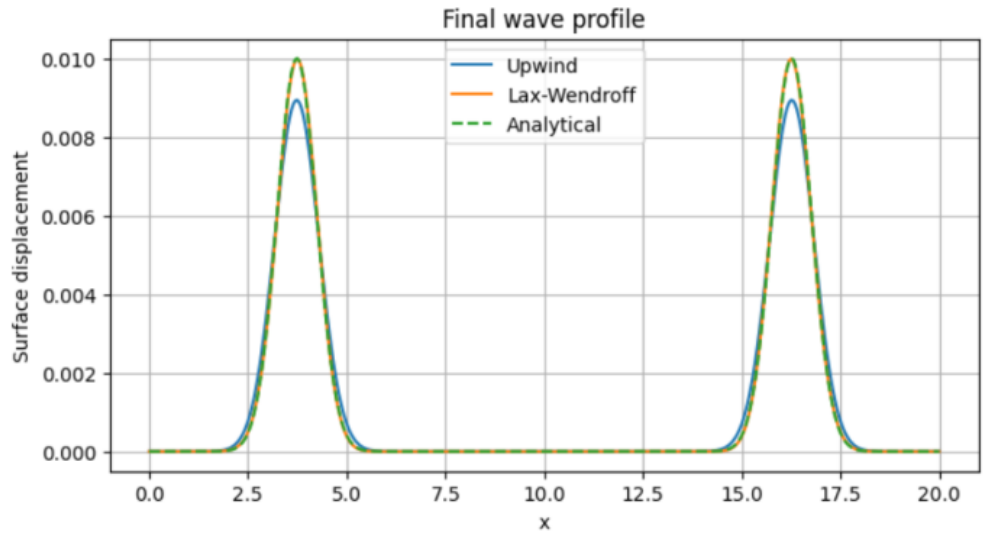
For CFL number 0.4

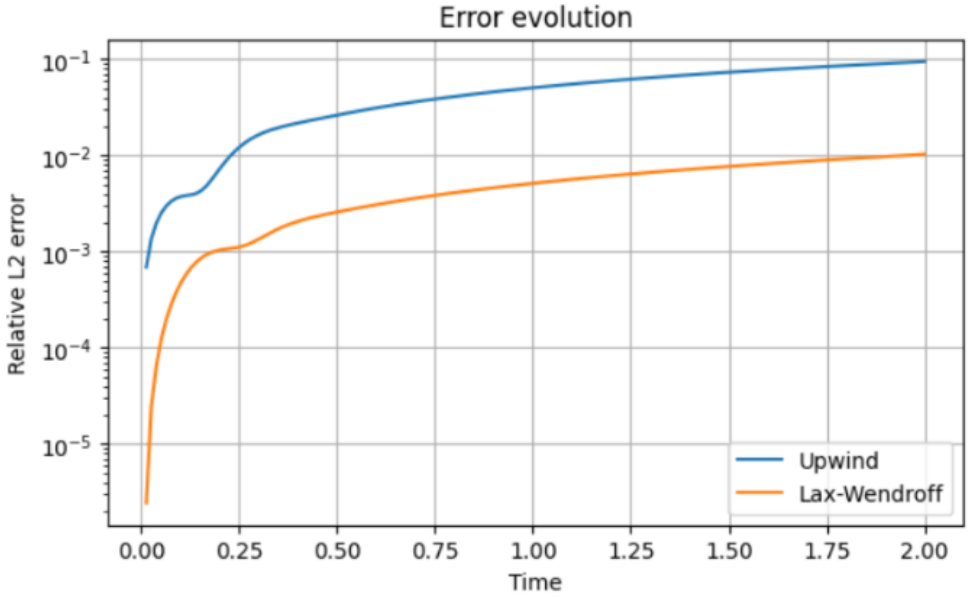


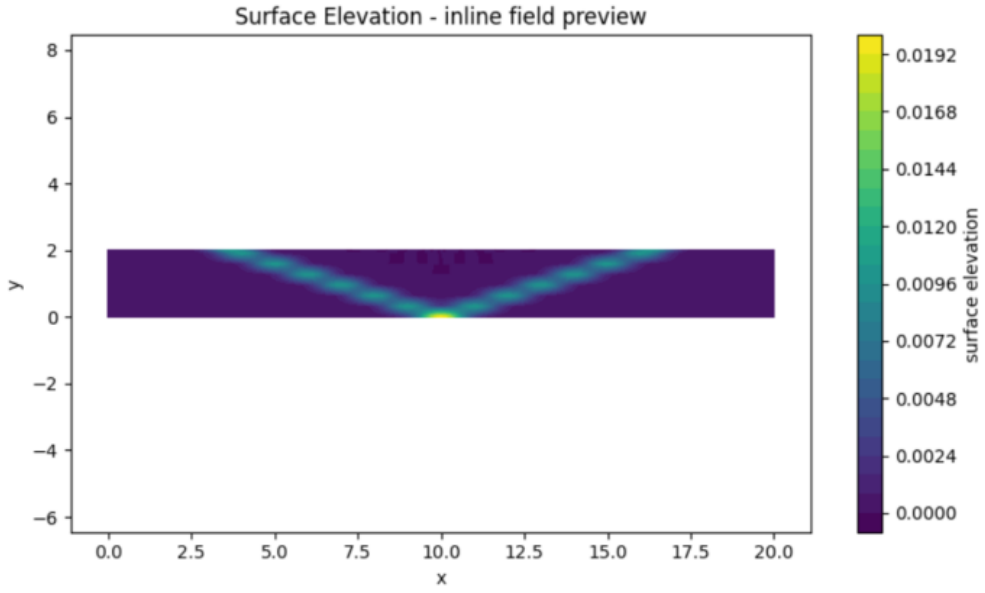




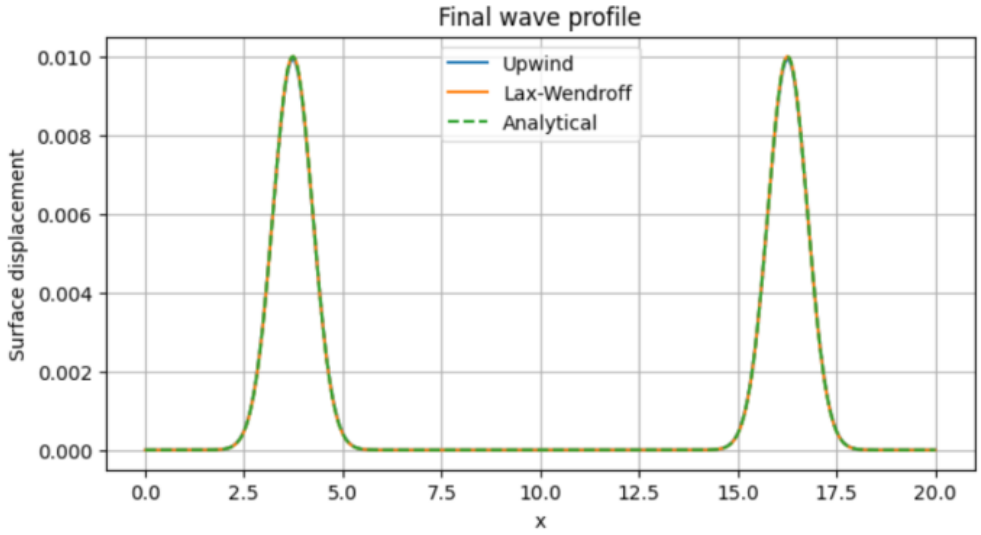
For CFL number 0.8

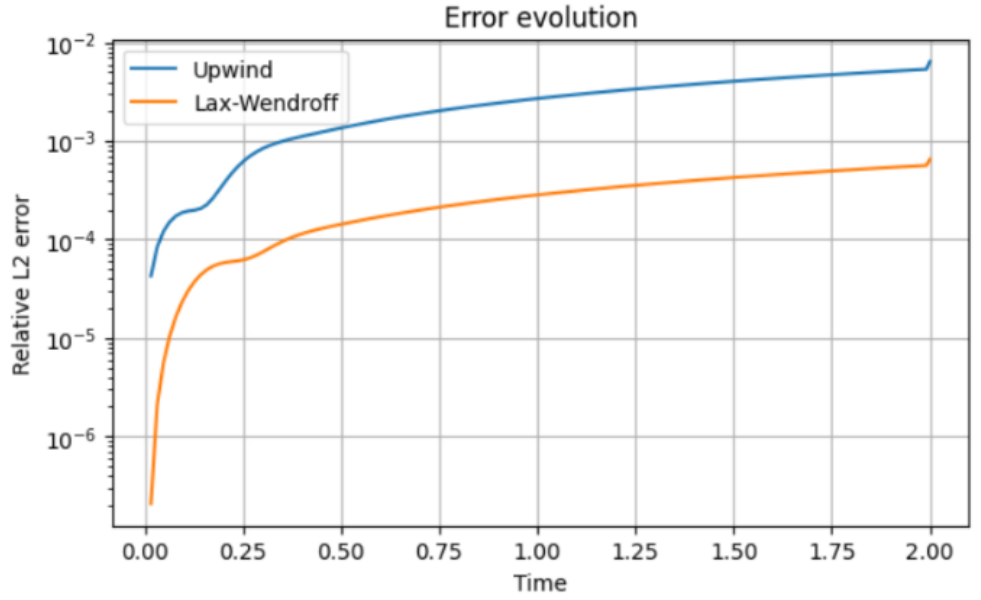


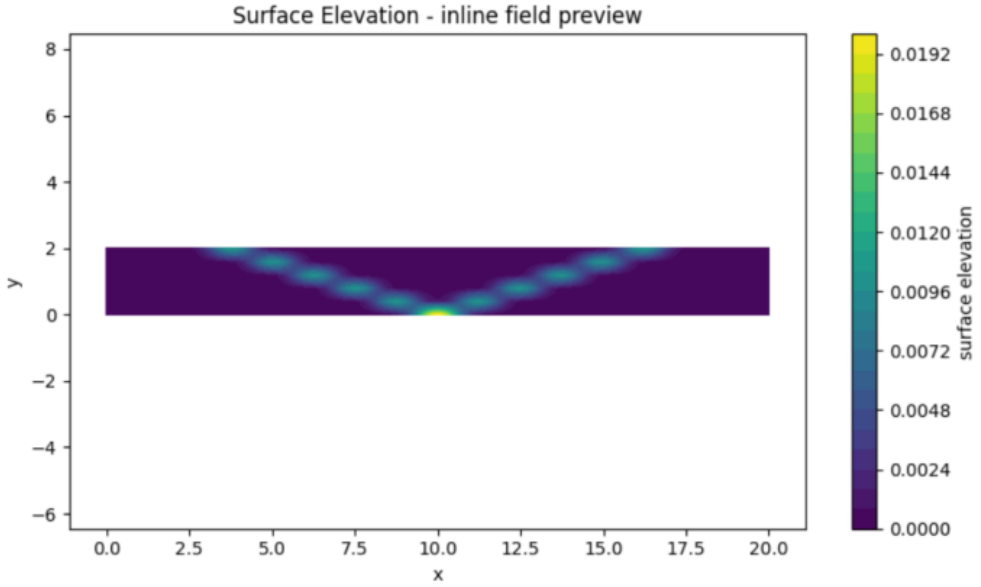

Error evolution
Relative L2 error
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
Time
Upwind
Lax-Wendroff


Surface Elevation - inline field preview
y
8
6
4
2
0
−2
−4
−6
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
x
0.0192
0.0168
0.0144
0.0120
0.0096
0.0072
0.0048
0.0024
0.0000
surface elevation

For CFL number 1.1







Results:

Parameter	**CFL = 0.4**	**CFL = 0.8**	**CFL = 1.1 (nominal)**
Pulse centres at t = 2 (x)	≈ 3.75 and 16.25	≈ 3.75 and 16.25	≈ 3.75 and 16.25
Analytical peak amplitude (each pulse)	≈ 0.0100	≈ 0.0100	≈ 0.0100
Upwind peak amplitude at t = 2	≈ 0.0076	≈ 0.0090	≈ 0.0100
Upwind amplitude loss	≈ 24%	≈ 10%	negligible
Lax–Wendroff peak amplitude at t = 2	≈ 0.0100	≈ 0.0100	≈ 0.0100
Upwind relative L^2 error at t = 2	$\approx 2.3 \times 10^{-1}$	$\approx 9.5 \times 10^{-2}$	$\approx 6 \times 10^{-3}$
Lax–Wendroff relative L^2 error at t = 2	$\approx 2.4 \times 10^{-2}$	$\approx 1.0 \times 10^{-2}$	$\approx 6 \times 10^{-4}$

Parameter	**CFL = 0.4**	**CFL = 0.8**	**CFL = 1.1 (nominal)**
Error ratio (upwind / Lax–Wendroff)	≈ 10	≈ 10	≈ 10
Phase error (peak position)	none visible	none visible	none visible
Spurious oscillations	none visible	none visible	none visible
Growth of error in time	smooth, monotonic	smooth, monotonic	smooth, monotonic
Mass conservation (conservative form, periodic domain)	conserved (by construction)	conserved (by construction)	conserved (by construction)

Conclusion:

The one-dimensional linearised shallow-water wave problem was solved on a periodic domain using the conservative first-order upwind scheme and the second-order Lax–Wendroff scheme, at CFL numbers of 0.4, 0.8 and 1.1, and the results were compared with the analytical solution. The initial Gaussian surface perturbation of amplitude 0.02 splits into two equal pulses of amplitude ≈ 0.010 that travel in opposite directions. At $t = 2$ the pulse centres are at $x \approx 3.75$ and $x \approx 16.25$, i.e. about 6.25 from the origin at $x = 10$, which corresponds to a wave speed of $c \approx 3.1$. This is consistent with $c = \sqrt{(gH)}$ for $g = 9.81$ and a depth of about 1. The surface-elevation field plots show the two symmetric wavefronts leaving the source in all three cases.

Numerical diffusion of the upwind scheme: The first-order upwind solution is strongly diffusive at low CFL. At CFL = 0.4 the pulse peak falls from 0.0100 to approximately 0.0076, an amplitude loss of about 24%, and the base of the pulse is visibly broadened. At CFL = 0.8 the peak is approximately 0.0090 (about 10% loss), and at the nominal CFL = 1.1 the upwind profile is indistinguishable from the analytical one. The corresponding relative L^2 errors at $t = 2$ are approximately 2.3×10^{-1}, 9.5×10^{-2} and 6×10^{-3}. The error grows steadily with time because the smearing accumulates as the pulses travel. This behaviour follows from the leading truncation error of upwind being a diffusive term, with an effective numerical viscosity of about $(c\Delta x/2)(1 - \mathrm{CFL})$, which becomes smaller as CFL approaches 1.

Accuracy of the Lax–Wendroff scheme: The second-order Lax–Wendroff solution retains a peak amplitude of approximately 0.0100 and overlaps the analytical profile in all three cases. Its relative L^2 errors at $t = 2$ are approximately 2.4×10^{-2}, 1.0×10^{-2} and 6×10^{-4}, about one order of magnitude smaller than those of upwind at every CFL value. The leading error of Lax–Wendroff

is dispersive and scales with $(1 - \mathrm{CFL}^2)$; the reduction of the error by a factor of about 2.4 between CFL = 0.4 and 0.8 agrees with the ratio $(1 - 0.4^2) / (1 - 0.8^2) \approx 2.3$. No phase error or trailing oscillations were visible in the plots. This is expected for a smooth Gaussian pulse resolved by roughly 25 grid points across its width, where dispersive errors remain small. Lax–Wendroff can produce oscillations near sharp gradients, but the present smooth case does not test this.

Effect of CFL number and stability: For both schemes the error decreases as the Courant number approaches unity, because at CFL = 1 the discrete update reproduces the exact characteristic shift. The explicit stability limit for both schemes is CFL ≤ 1. At the nominal CFL = 1.1, however, the plots show no growth of oscillations or error: both error histories continue the downward trend seen from CFL = 0.4 to 0.8. This indicates that the solver kept the effective Courant number at or just below unity, for example by limiting the time step, rather than running at 1.1. The nominal CFL = 1.1 case therefore does not demonstrate the instability that theory predicts for CFL > 1, which would appear as exponentially growing oscillations. Demonstrating the stability limit would require a run without a time-step limiter.

Conservation and overall comparison: Both schemes are written in conservative form on a periodic domain, so the total surface elevation (mass) is conserved by construction. The amplitude loss of the upwind solution is therefore a redistribution of the same volume into lower, wider pulses, not a loss of mass. Overall, Lax–Wendroff is about ten times more accurate than upwind at the same CFL, with no visible dispersion at this resolution, while upwind is robust but strongly diffusive unless CFL is close to 1. The accuracy of both schemes improves as CFL approaches 1, and the stability bound CFL ≤ 1 governs the choice of time step for these explicit methods.